

MATHEMATICS DEPARTMENT

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GENERAL MATHEMATICS

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STRENGTHENED SENIOR HIGH SCHOOL CURRICULUM



OPENING PRAYER



MATH CHECK



Make your self ready

Align your chairs properly

Take away all the scraps

Have your seats

Attendance





KAMUSTA KA?

MATHEMATICS DEPARTMENT



RECAP

STRENGTHENED SENIOR HIGH SCHOOL CURRICULUM



RECAP

Key Measures in Statistics

Let's explore the basic measures used to describe a set of data!

Mean ($\bar{x}$ or μ)

The average sum of all values divided by the number of data points (n).

$$\bar{x} = \frac{\sum x}{n}$$

Example:

Data: 4, 7, 10, 13, 16

$$\begin{aligned}\bar{x} &= \frac{4 + 7 + 10 + 13 + 16}{5} \\ &= \frac{50}{5} = 10\end{aligned}$$



Median

The middle value when data is arranged in ascending order. (Calculators generally do not have a direct button for raw data median on older models).

3 | 5 | 7 | 9 | 12

Median = 7

(middle value)

Mode

The value that appears most frequently. (Scientific calculators do not compute the mode automatically).

Example:

Data: 2, 4, 4, 6, 8, 10

Mode = 4

(most frequent)





Range

The difference between the highest and lowest values ($R = x_{\max} - x_{\min}$).

Example:

Data: 5, 8, 12, 17, 20

$$R = 20 - 5 = 15$$



Better
Data
=
Better
Decisions

Variance (s^2 or $\bar{\sigma}^2$)

The average of the squared differences from the mean.

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} \quad \text{or} \quad \sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

(sample) (population)

Example:

Data: 2, 4, 6, 8, 10

Mean ($\bar{x}$) = 6

Variance (s^2) = 10

→ (calculated using the formula)

Standard Deviation (s or $\bar{\sigma}$)

The square root of the variance,
measuring data spread.

$$s = \sqrt{s^2}$$

Example:

If $s^2 = 10$,

$$s = \sqrt{10} \approx 3.16$$


In summary:

Mean
average

Median
middle value

Mode
most frequent

Range
spread (max – min)

Variance
average squared
difference

Standard Deviation
spread (in original units)

LESSON OBJECTIVES

- 🎯 Identify the appropriate spreadsheet functions (AVERAGE, MEDIAN, MODE.SNGL, STDEV.S, VAR.S, MAX, MIN) and scientific calculator workflows for computing central tendency and variability.;
- 🎯 Calculate measures of central tendency (mean, median, mode) and variability (range, variance, standard deviation) using digital tools (MS Excel / Google Sheets or Scientific Calculators) for a given dataset; and

LESSON OBJECTIVES

🎯 Demonstrate digital integrity and meticulousness when encoding and computing data using technology.



Activity 1:

The Human Spreadsheet and the Data Crunch

Data:

2, 4, 4, 7, 8 hours of study.



2 minutes



Think. Calculate.
Share!

Instructions:

1

Look at the data:
2, 4, 4, 7, 8 hours of study.

2

Work with a partner.

3

In 2 minutes, find the:

- Mean
- Median
- Mode
- Range

4

Use mental math or scratch paper to solve.

5

Be ready to share your answers and solution with the class.



Small numbers.
Big insights!

If the Department of Education division office requires us to submit the **mean, median, standard deviation, and variance** of these **150** scores within the next **five minutes** to identify which schools need urgent intervention, how would we calculate this manually, and what challenges would we face?

150 SCORES				
78	85	92	67	74
88	76	90	83	71
95	62	80	77	69
... (total of 150)				

Calculate:

- Mean
- Median
- Standard Deviation
- Variance



Possible Challenges: ?

- Time-consuming?
- Tedious calculations?
- Risk of errors?
- Handling large data?
- What formulas should we use?

How would we calculate this manually, and what challenges would we face?



Computational Integration via Tech Platforms

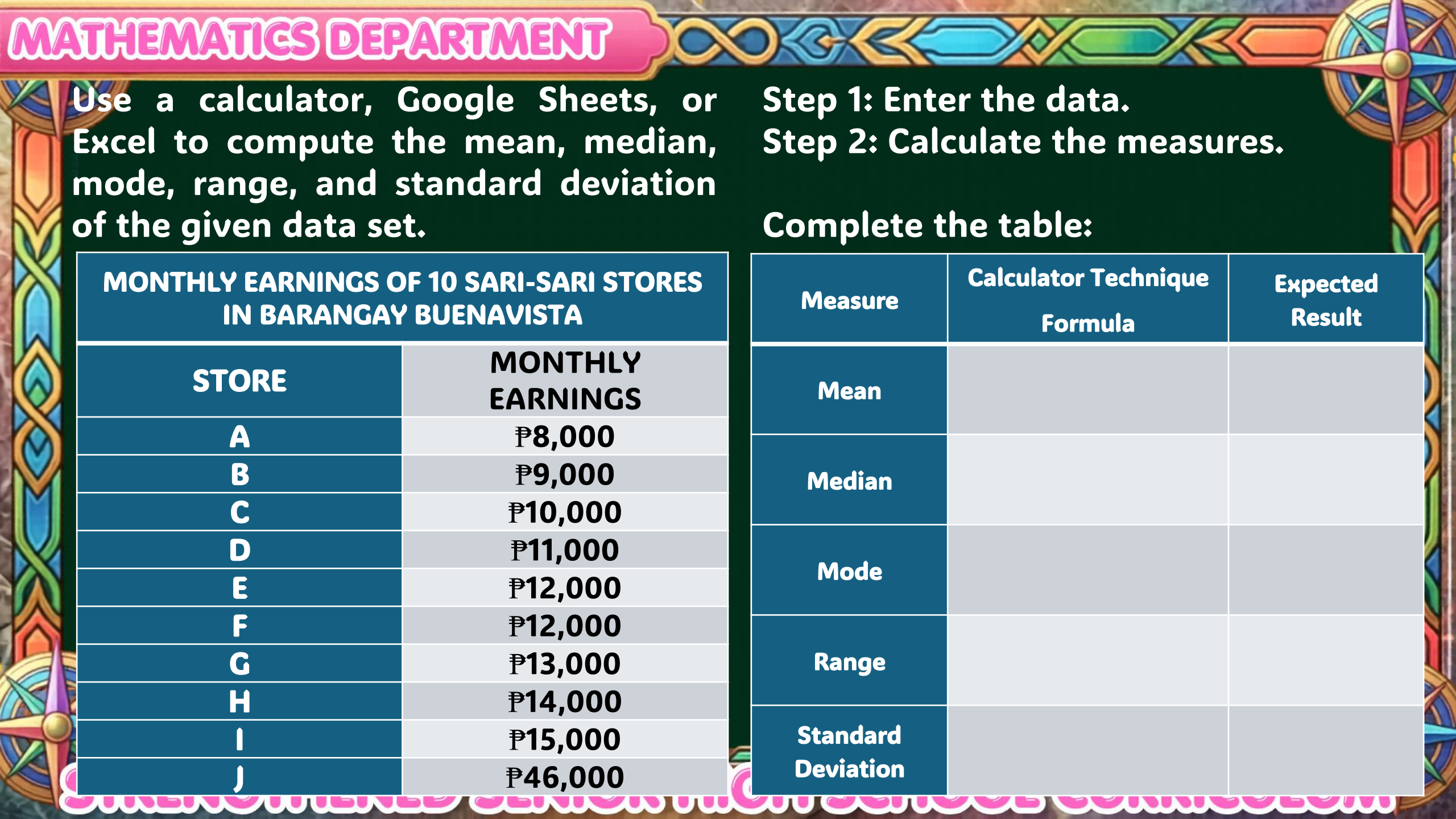
**PRESENTED BY:
MELQUI JOHN B. ARANDIA
Teacher V**

Activity 2:

TECH-ASSISTED DATA CALCULATION LAB



Scenario: "Imagine that the barangay wants to understand the financial situation of small sari-sari stores. We have the monthly earnings of 10 stores. Instead of calculating everything manually, we will use technology to help us analyze the data quickly."



MATHEMATICS DEPARTMENT

Use a calculator, Google Sheets, or Excel to compute the mean, median, mode, range, and standard deviation of the given data set.

Step 1: Enter the data.

Step 2: Calculate the measures.

Complete the table:

MONTHLY EARNINGS OF 10 SARI-SARI STORES IN BARANGAY BUENAVISTA	
STORE	MONTHLY EARNINGS
A	₱8,000
B	₱9,000
C	₱10,000
D	₱11,000
E	₱12,000
F	₱12,000
G	₱13,000
H	₱14,000
I	₱15,000
J	₱46,000

Measure	Calculator Technique Formula	Expected Result
Mean		
Median		
Mode		
Range		
Standard Deviation		

Direction: Find the MEAN, STANDARD DEVIATION and VARIANCE

Calculator Technique:

Set up your Calculator



Entering the given (Probability Distribution)



Finding the Answers



MONTHLY EARNINGS OF 10 SARI-SARI STORES IN BARANGAY BUENAVISTA	
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H	₱14,000
I	₱15,000
J	₱46,000



LET'S ANALYZE

The standard deviation is approximately ₱11,447. What does this tell us about the monthly earnings? If you were planning to provide a loan or financial assistance to these stores, would it be reasonable to assume that all stores have similar earning capacity?"





If median income is ₱12,000 but mean income is ₱25,000, what does this reveal about economic distribution in the barangay?

Why is standard deviation preferred over range for comprehensive risk assessment?





LET'S SUM IT UP!

Mean tells us the average.

Median tells us the middle value.

Mode tells us the most frequently occurring value.

Range tells us the distance between the highest and lowest values.

Standard deviation tells us how spread out the values are around the mean.

FUNDAMENTAL STATISTICAL MEASURES



Mean ($\bar{x}$ or μ): The average sum of all values divided by the number of data points (n).

$$\bar{x} = \frac{\sum x}{n}$$

sum of values



Median: The middle value when data is arranged in ascending order.



Mode: The value that appears most frequently.



Range: The difference between the highest and lowest values.

$$R = x_{\max} - x_{\min}$$

Variance (s^2 or σ^2): The average of the squared differences from the mean.

$$s^2$$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

(sample)

$$\sigma^2$$

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

(population)



Standard Deviation (s or σ): The square root of the variance, measuring data spread.

$$s = \sqrt{s^2} \quad \sigma = \sqrt{\sigma^2}$$

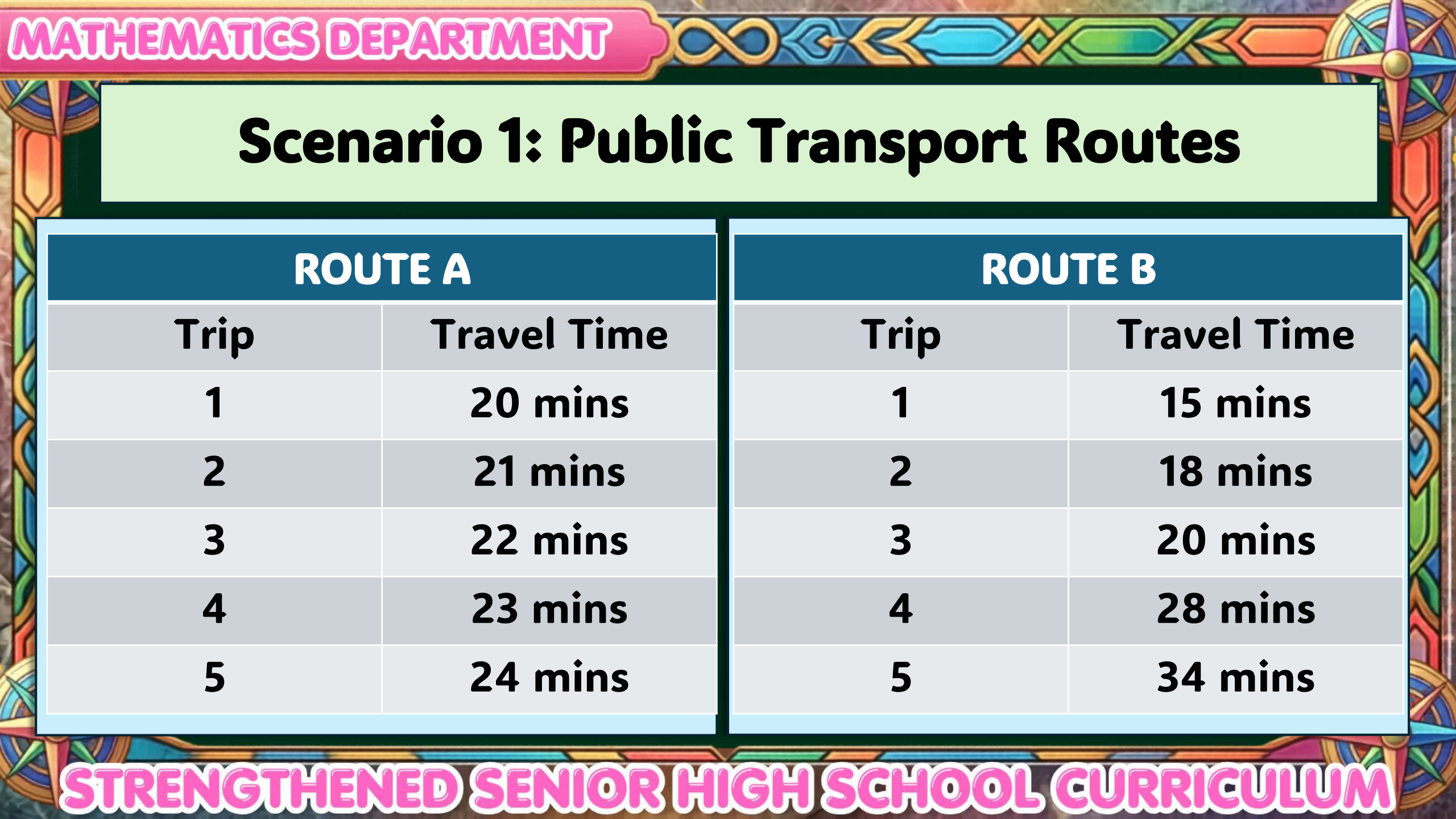
sqrt of variance



**LET'S
APPLY**

Activity 3: THE BARANGAY DATA STORYTELLER CHALLENGE

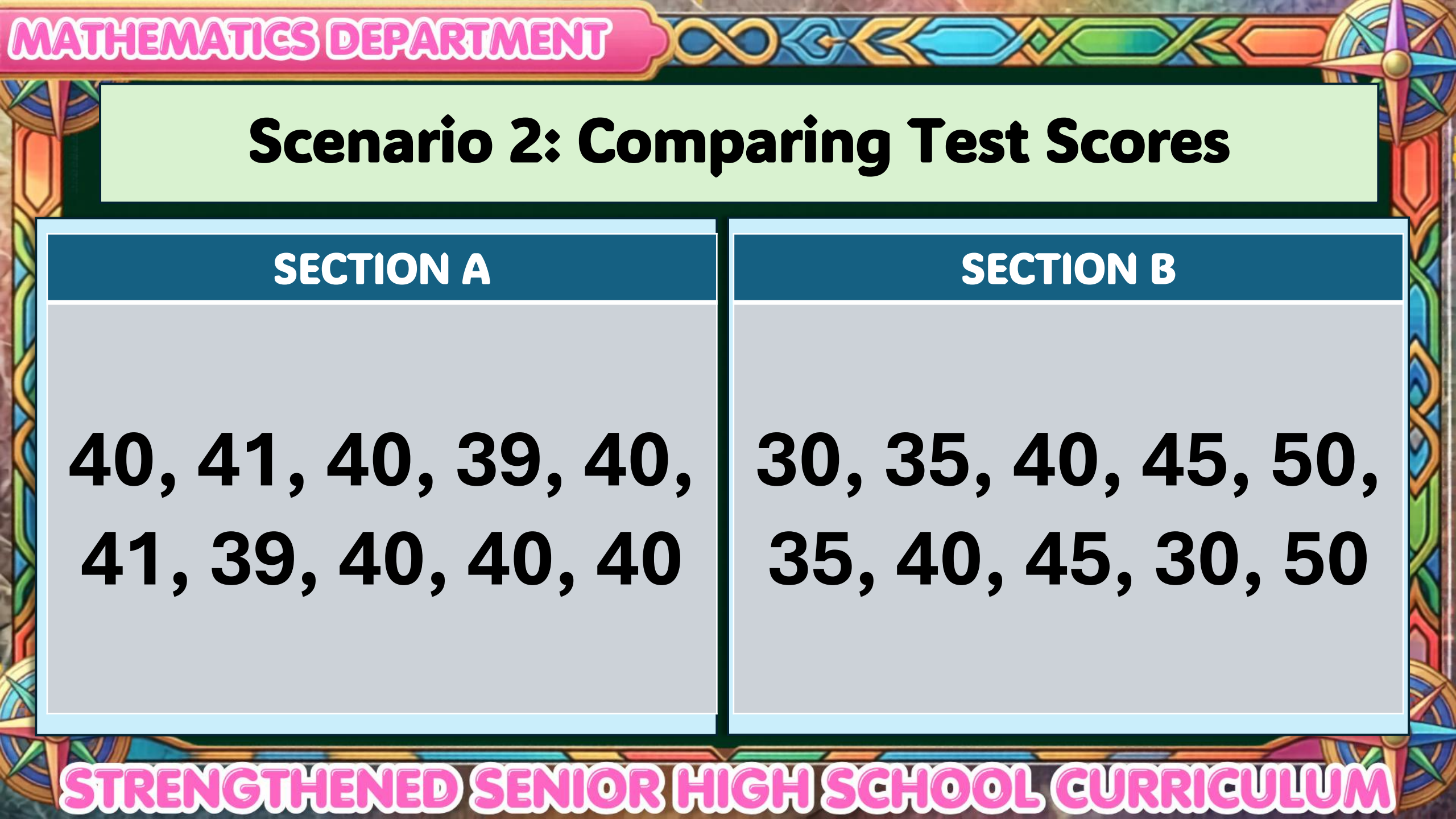
Groups receive real-world scenarios (e.g., Evaluating two public transport routes, comparing test consistency between two STEM sections). They compute all statistical measures and present a 2-minute "Data Story Pitch" recommending concrete administrative action based on statistical findings.



Scenario 1: Public Transport Routes

ROUTE A	
Trip	Travel Time
1	20 mins
2	21 mins
3	22 mins
4	23 mins
5	24 mins

ROUTE B	
Trip	Travel Time
1	15 mins
2	18 mins
3	20 mins
4	28 mins
5	34 mins



Scenario 2: Comparing Test Scores

SECTION A
40, 41, 40, 39, 40, 41, 39, 40, 40, 40

SECTION B
30, 35, 40, 45, 50, 35, 40, 45, 30, 50

2-MINUTE "DATA STORY PITCH"

Each group presents its findings using this simple structure:

- 1. DATA** "Our dataset is about _____."
- 2. CENTER** "The mean is _____ and the median is _____."
- 3. SPREAD** "The standard deviation is _____ and the range is _____."
- 4. INTERPRETATION** "This means that _____."
- 5. ACTION** "Based on the data, we suggest that the barangay/school administration should _____."

A yellow rectangular graphic with a scalloped border, a vertical line of ten circular punch holes on the left side, and a barcode on the far left. The text "EXIT TICKET" is centered in the middle of the graphic.

EXIT TICKET

⇒ EXIT TICKET

Which spreadsheet formula correctly calculates the sample standard deviation for data in cells B2 to B20?

- A. =STDEV(B2:B20)**
- B. =STDEV.S(B2:B20)**
- C. =VARIANCE(B2:B20)**
- D. =AVERAGE.DEV(B2:B20)**



⇒ EXIT TICKET

2. In MS Excel or Google Sheets, what formula sequence finds the Range of data in cells A1:A50?

- A. =RANGE(A1:A50)
- B. =MAX(A1:A50) - MIN(A1:A50)
- C. =SUBTRACT(A1:A50)
- D. =SUM(A1:A50)/50



 EXIT TICKET

3. What happens to the `=AVERAGE()` result in a dataset if a typo adds an extra zero to a number?

- A. The mean remains unaffected.
- B. The mean decreases significantly.
- C. The mean increases significantly because it is sensitive to extreme values.
- D. The spreadsheet displays a `#VALUE!` error automatically.



 EXIT TICKET

A student inputs 10 test scores into a spreadsheet and applies `=MEDIAN(C1:C10)`. The function outputs 85. What does this value mean?

- A. 85 is the most frequent score.
- B. The average of all scores is 85.
- C. Exactly half of the test scores are at or below 85, and half are at or above 85.
- D. The difference between highest and lowest score is 85.



⇒ EXIT TICKET

5. If a calculated sample standard deviation (s) using tech is 0.5 for group A and 4.8 for group B, what does this indicate?

- A. Group A has higher variability than Group B.**
- B. Group B scores are much more spread out from their mean than Group A scores.**
- C. Group A has a higher mean than Group B.**
- D. Group B contains invalid data entry.**



WAYS FORWARD

Tech Practice: Data Cleaning Exercise

Students will download a sample messy dataset provided by the teacher (containing intentional formatting inconsistencies and typos). They will use spreadsheet functions to clean the dataset, re-calculate the correct measures of central tendency and variability, and submit the corrected spreadsheet file.

SESSION DEVELOPER

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